

Commercial Model to Increase Sales through Mixed Methodologies in an Automotive Retailer

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ABSTRACT

In recent years, the digital channel of companies in the automotive retail industry has experienced considerable growth, reflected in the percentage of sales of this channel with respect to total sales. Therefore, companies that are not aligned with this growth, are those that have inefficiencies in their processes of attention to leads. Processes that consist of different procedures and essential operations that are not considered, nor are they applied efficiently, ignoring the magnitude of the impact on the company's sales. This research focuses on analyzing 4 tools to obtain a new commercial model that optimizes and improves the low conversion of leads into sales, due to inefficient lead handling processes, identified as the main problems within the company studied. The model makes use of tools such as: demand management based on Forecasting, optimization of the process as such through BPM, improvement of the CRM database through Poka Yoke, and improvement of salespeople skills under a profile standardization approach, resulting in a 15.73% improvement in the company's net profit, providing an effective commercial model with the capacity to be replicated in other companies.

CCS CONCEPTS

• Applied computing; • Operations research; • Industry and manufacturing; • Supply chain management;

KEYWORDS

Business Process Management, Forecasting, Poka Yoke, Profile Standardization

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1 INTRODUCTION

Online sales are becoming more and more frequent in all sectors; however, the automotive retail industry is still skeptical about the change, mostly by the brands themselves. Currently, there are very few companies that really bet on a 100% digitized sales process in Latin America and in a country like Peru this figure is reduced to only 3 brands [1]. In this area, e-commerce in Peru moved 6 billion dollars in 2020, of which the automotive retail industry had practically no participation [2]. As a result of the pandemic, only 125,090 units of light and heavy vehicles were sold in 2020, representing a 26% drop compared to 2019, the lowest figures since 2010 [3]. While several industries took advantage of the pandemic to venture into e-commerce and digitize their sales processes, the automotive industry has failed to adapt to this sudden and abrupt change. A Euromonitor International study conducted for Google indicates that by 2025, e-commerce in Peru will grow up to 110% [4]. This research highlights the importance of increasing the conversion rate of digital leads using a methodological approach.

The problem identified according to the literature leads us to the case study of an automotive retail company where the main problem are the loss of digital leads, which may be due to an excess volume of customers, repeat customers, lack of capacity by salespeople, among others. This problem has also been identified in various research studies in different countries. For example, research conducted in a New Zealand automotive retail company in 2019, based on the adoption and implementation of e-commerce in the automotive industry, determined that 80% of the leads were derived through the company's website [5]. This issue has also been identified in a study conducted by Capgemini Consulting, which found that 60% of consumers in the United States are willing to buy a new car online, while in China and Germany the percentage of consumers who would be willing to make their purchase 100% digitally increases by 77% [6]. In another research conducted by Ruler Analytics, it was found that the global average conversion rate of leads coming from digital channels was equal to 2%, with an exponential growth trend for the next five years [7]. The above-mentioned shows that companies in the automotive retail sector in Peru do not have

adequate processes to take full advantage of the capture of leads that are derived through digital channels, so it should continue to investigate new engineering solutions to this problem.

In this context, it is necessary that automotive retail companies in Peru can take full advantage of leads from digital channels. For this purpose, a case study was chosen to reflect the problem of the low conversion rate of digital leads in the sector. The problems presented were related to poor planning when hiring the sales force, poor definition of digital strategy, location infrastructure, training structure, contract structure, among others. In this sense, to solve the problems described above, a commercial model of improvement was developed combining methodologies, starting with the correct management of the demand for leads with Forecasting, passing to the application of BPM, for the correction of bottlenecks and optimization of process time, Poka Yoke to correct the management of the company's database and finally, Standardization of Profiles, since it is the digital salespeople who manage the entire process of attention of the company's digital leads. This model was developed based on the success cases that presented similar problems to the case study and is intended to satisfy the need to solve the sector's problems and contribute to the scientific community. The present research offers a business model of digital transformation focused on increasing the conversion rate of digital leads.

The scientific articles reviewed contain little information on digital transformation business models for automotive companies, especially in Peru. For that reason, the need to conduct the present research arises. The scientific article presented below is divided into seven parts which are Introduction, State of the Art, Methodology, Contribution, Validation, Acknowledgements and Bibliography.

2 STATE OF THE ART

2.1 Business Model to Increase Sale in the Automotive Industry

The first category shows that most business models in the automotive retail industry are based on strategic planning [8] [9]. This is defined as the deliberate and disciplined approach to generate decisions and actions with the objective of guiding the organization to meet its sales objectives [10]. Also, the analysis of the role of social networks at the time when consumers make the decision to buy a car is highlighted. This is because social networks play an important role in promoting effective customer support information. These are an interactive platform where customers have an open voice opinion of each product or service and a greater veracity within all traditional media [11]. In addition, several articles cover the topic of "service dominant", which is based on the creation of service through the application of competencies by individuals, thus demonstrating that, although managers can apply the "service dominant" methodology and its principles as part of their personal mentality, they will obtain better benefits for the company if they apply it directly to the sales force [12].

2.2 Business Process Management (BPM) in the Automotive Industry

Organizations must monitor, measure, and improve their processes if they want to survive in today's dynamic market [10]. Therefore,

in recent decades, there has been an increased interest in business processes. Due to the need to streamline business operations, consolidate organizations and reduce costs, highlighting the importance and value of the process as the basic unit of business value within a corporation [13] [14]. In many business organizations there is interest in BPM, as reflected in the report entitled: "The State of Business Process Management" published in 2018, which shows interesting data such as: 93% of the organizations in the survey are involved in multiple process improvement projects; 79% said they were using process software tools to model the processes they were analyzing and planning to change [15] [16]. In addition, the ease of use, due to the support of tools that allow BPM users to have greater flexibility.

2.3 Lean Six Sigma (Poka Yoke) in the Automotive Industry

According to the studies found, the Poka Yoke tool is mainly used in the automotive industry as a method to reduce the percentage of human errors in the production process and, therefore, reduce costs [10]. However, it can also be used in information systems to optimize them and generate greater efficiency in the software. Through Poka Yoke ISD ("Information system design"), unwanted data can be eliminated in operations that require ERP systems for their operation. Traditional Poka-Yoke ED relies primarily on physical characteristics for error detection and its application has been limited mainly to production processes. In contrast, Poka-Yoke LSD has no such limitation and, therefore, can be implemented in both manufacturing and service processes [11].

2.4 Forecasting in the Automotive Industry

Companies in the automotive industry are facing a highly competitive environment in constant growth, as the current market increasingly presents greater access to tools, due to the constant technological development. On the other hand, according to in 2019, 20 countries exceeded in 1% the market share of electric vehicles [19], which highlights the importance of demand management, due to an overspending of resources by many companies in the automotive industry. Also, an inadequate estimation of demand can lead to a loss of sales or excess inventory, increasing the company's costs [20]. Therefore, being able to meet market demand assertively becomes a key factor to maintain competitiveness in the medium and long term. Since the entire operations and supply chain of any company depend on a correctly forecasted demand [18] [21].

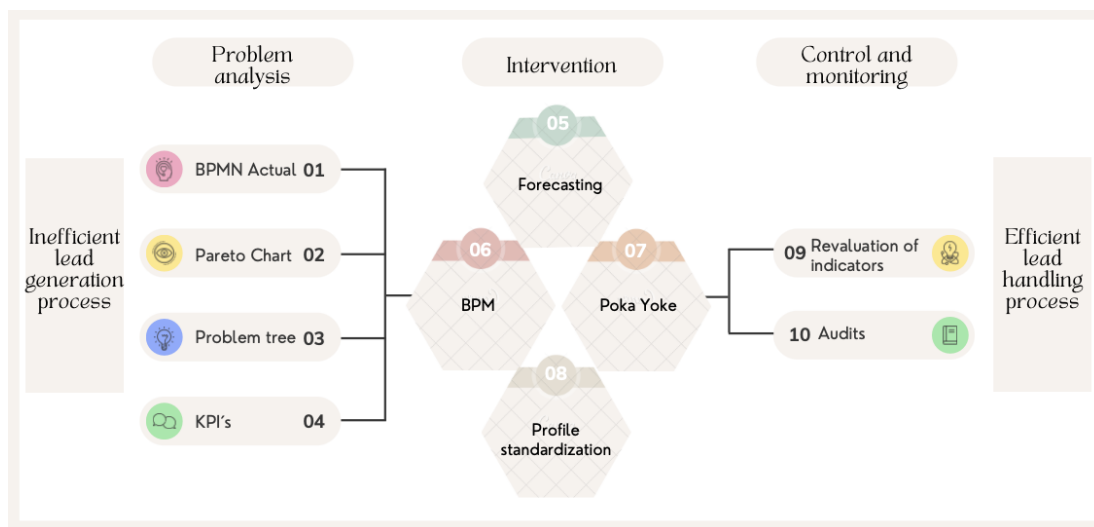
Table 1 below shows the authors who have contributed the most to identify the tools required for the proposed model:

3 PROPOSED MODEL

Figure 1 shows the innovative proposal, which focuses on redesigning the lead attention process through a commercial approach based on technological improvement. Unlike the other models found, the model focuses on optimizing and improving the level of attention of leads of less than 6 hours by the sales force, with the objective of streamlining the entire process, from lead capture to the conclusion of the same with the aim of increasing the conversion rate, for this it is necessary to increase the efficiency of workers which is the critical point of the sales process where more potential customers

Table 1: Comparative Matrix of the Objectives of the Problem vs. the State of The Art

Authors/ Causes	Excess customer volumen	Extended process time	Low skill level	Repeat customers
[8]			Profile Standardization	
[12]			Confirmatory Factor Analysis (CFA)	
[13]		Business Process Management (BPM)		
[14]		Business Process Management (BPM)	Business Process Management (BPM)	
[16]		Business Process Management (BPM)		
[17]				Poka Yoke
[18]				Poka Yoke
[19]	Forecasting			
[20]	Forecasting			
Proposal	Forecasting	Business Process Management (BPM)	Profile standardization	Poka Yoke

**Figure 1: Proposed Improvement Model**

are lost. Forecasting tools will be used to solve the excess volume of customers, BPM to lengthen the lead processing time, standardization of profiles for the low level of skills of salespeople and poka yoke to solve the repetition of customers in the CRM.

3.1 Model Components

The model is divided into three components, as mentioned below.

3.1.1 Component 1: Problem Analysis. This component consists of all the activities prior to the implementation of the proposed model. First, a previous analysis of the current situation and a BPMN was performed, to understand more specifically the lead attention process and to determine the efficiency of each step of the process.

Secondly, once the mapping was done and possible problems and opportunities for improvement were identified, a pareto chart was made to identify the problems and prioritize. Third, a problem tree was drawn up to find the root causes of the problems. Once the root causes were identified, KPI's were established to measure the root causes of the current situation and find the variation once the model tools were implemented.

3.1.2 Component 2: Intervention. The second component is based on the implementation of the tools described in the model. The first tool implemented is Forecasting, which aims at the efficient management of automobile demand.

Table 2: Comparison of Indicators

Indicator	Formula	Use
% Digital leads	$(\text{Digital leads} / \text{Total leads}) * 100$	Number of leads received through digital channels over the total number of leads received through all channels.
% Leads served in less than 6 hours	$(\text{Leads served under 6 hours} / \text{Total leads}) * 100$	Number of digital leads contacted in less than 6 hours over the total number of leads.
% Digital lead conversion rate	$(\text{Digital sales} / \text{Digital leads}) * 100$	Digital sales over total digital leads received.
% Repeat customers	$(\text{Repeat customers} / \text{Total customers}) * 100$	Repeat customers in the CRM (Salesforce) over total number of customers assigned to each customer in the CRM (Salesforce).

After forecasting the demand for automobiles by means of Forecasting, a BPM of the processes is carried out to subsequently make the necessary changes in the inefficient processes.

For the evaluation of the processes, two criteria were considered, the first one the amount of time of the process, since this way a focus is given to reduce the time of the longest processes, and the second one the processes with technological failures, since this way an improvement in the tools used is achieved. Subsequently, a control is performed with respect to repeat customers, implementing the Poka Yoke methodology, which aims to optimize the use of CRM. This made it possible to optimize the database used by the salespeople, following the principles focused on a culture of prevention.

Finally, a profile standardization was carried out to improve the low level of skills of the salespeople. In the application of profile standardization, in step 1, the salespeople in charge of lead management were considered. As a second point, the design of variables to be measured for the validity of the application of the tool.

3.1.3 Component 3: Control and Monitoring. The last component consists of two activities once the implementation has been completed. These consist of a review of the KPI's established prior to implementation, to identify the impact of the changes generated by the model tools used. Finally, audits as a parameter for continuous improvement, since once the variation of the impact of the tools was measured, a diagnosis was made to identify opportunities for improvement and generate a virtuous circle of efficiency in the processes of attention to the leads.

Finally, with the results obtained from the simulation, a comparison table will be made with the initial information, and improvements of preventive and corrective actions related to the proposed tools will be proposed. The results are controlled by the table of indicators presented in Table 2.

4 VALIDATION

4.1 Initial Diagnosis

The automotive industry sells about 112,181 cars per year, generating 400,000 direct jobs and about 800,000 indirect jobs, absorbing about 7% of the Economically Active Population (EAP), according to the Peruvian Automotive Association (AAP). The model proposed in this research aims to study an automotive retail company in Lima Peru. The company has 17 locations, one of which was chosen for the study. Regarding the conversion rate of digital leads, a study conducted by Ruler Analytics found that the automotive industry

has a conversion rate of digital leads to sales of 2% worldwide [7]. The company's digital lead conversion rate was calculated over the year 2021. It was found that the company over 9 months of study obtained an average of 0.9%, This indicator generates a technical gap of 1.1%, which shows an opportunity for improvement.

Four main causes were found that cause the low number of leads in the process, these are the excess volume of customers representing 38.8%, extended process time 31.5%, low skill level 27.36% and repeat customers representing 2.35%. Figure 2 shows the problem tree that summarizes the diagnosis made in the case study:

4.2 Validation of the Design and Comparison with the Initial Diagnosis

To validate the design, it was necessary to cover three different components. The first component is focused on identifying the root causes of the loss of digital leads through a problem tree, where it was obtained that 38.80% is due to the excess volume of customers in the system, 31.5% of the time is due to extended processing time, 27.36% due to the low level of skills and 2.35% is due to repeat customers in the CRM.

In component two, the tools described in the model were implemented, starting with forecasting, where the demand projection and the company's current market share were obtained based on registrations and deliveries. Then a BPM was performed to evaluate the process time; subsequently, the Poka Yoke tool was implemented to eliminate repeat customers from the system and optimize the use of CRM. Finally, a profile standardization was carried out with the objective of improving the low CRM skill level of the EDVs.

The third component is based on a re-evaluation of indicators with the objective of measuring the KPIs previously established in the implementation.

4.3 Improvement - Simulation Proposal

The BMP tool was used to carry out a simulation prior to the implementations, from which the forecast was taken to analyze the demand together with the number of quotes, which gave us the following data: the time it takes to make a complete quote, the total time employed for one and the total of the quotations. Then, the Poka Yoke methodology was used where the steps that did not add value to the process were eliminated and others that added value were added, such as the inclusion of a pre-filter by the contact center. Finally, a standardization of profiles was carried out to keep

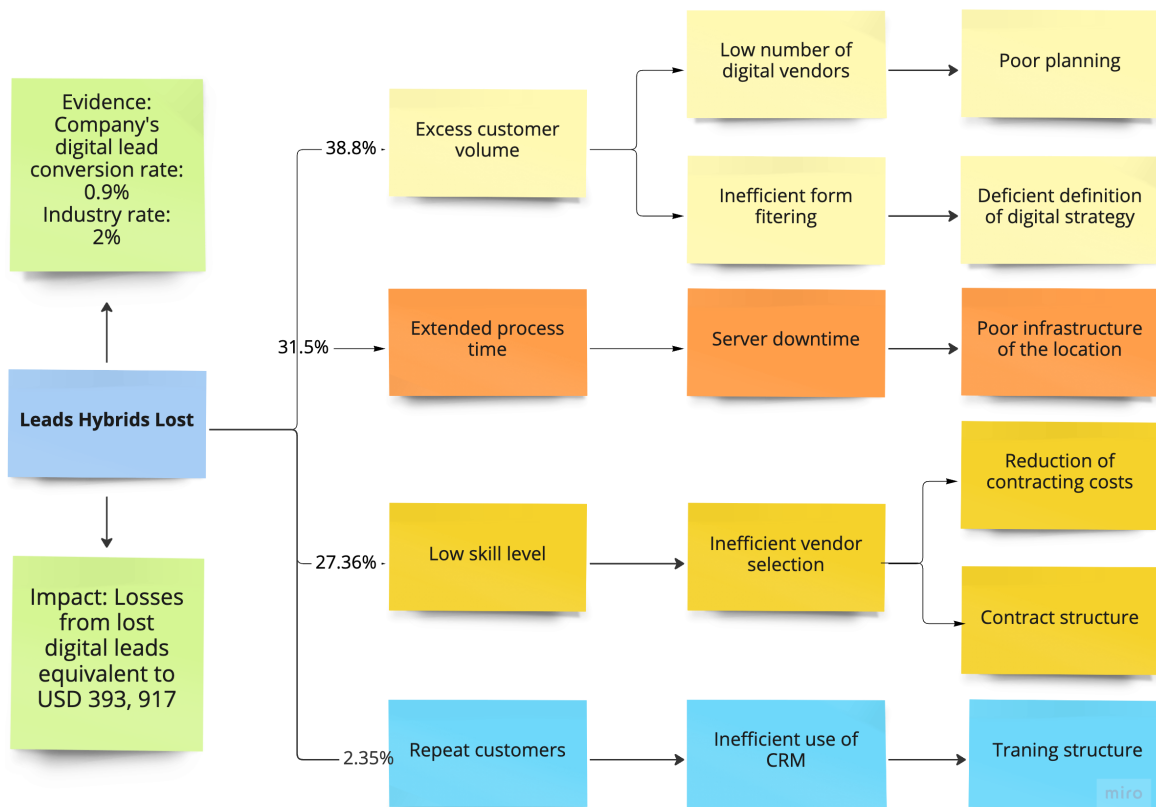


Figure 2: Problem Tree

a constant control of the sellers with the results that the simulation produced.

As a validation tool, a simulation was performed in the Arena software, with the purpose of validating the viability of the previously mentioned tools in the commercial model, confirming the efficiency of its operation.

This section will detail the simulation conditions, such as: system scope, input variables, sample size (runs), entities and restrictions, implementation areas, as well as the orders involved in the analyzed test period.

For the first scenario, a simulation was performed for the attention of leads, which arrive in time intervals that conform to an exponential distribution with a mean of 14 minutes. These are attended by 6 executives in 25% of the cases, 8 in 45% and 10 in 30% of the cases. They are then filtered by the contact center. Every 10 minutes the contact center team asks the sales manager if the number of leads received by the executives is less than or equal to 6. When this condition is met, the contact center team takes 20 seconds to assign the leads to the respective executives and assigns a maximum of 15 leads. The first group of leads enters at the beginning of the simulation. Once inside, the leads take 2 minutes to be chosen by each executive and are served with an even distribution of 4 to 10 minutes. At the end of the attention, the leads go to the quotation process. Figure 3 shows the simulation of the improved customer service process.

The second scenario covers the quotation process, where customers arrive at the store at time intervals that conform to an exponential distribution with a mean of 20 minutes. They take between 5 to 30 minutes to choose a brand. Once the brand has been chosen, the security guard assigns each customer an executive. There are 10 executives for customer service, with equal probability of being assigned. The time to assign the number has a uniform distribution between 1 and 5 minutes depending on the availability of the executives. The time of attention with any executive has a uniform distribution between 3 and 5 minutes. At the end of the attention with the executive, the customer leaves. Figure 4 shows the simulation of the improved quotation process.

Figure 5 shows the indicator board of the proposed molding and the results of both the diagnosis and the simulation results.

5 DISCUSSION

The proposed model based on increasing sales in an automotive company using lean six sigma, BPM, and Forecasting tools can be adapted to other companies in the retail sector that present similar problems related to excessive customer volume, extended processing time, low skill level, and repeat customers in the CRM. The results of the application of the model are shown in Figure 4.

When examining the results obtained, the indicator with the most significant variation was the digital lead conversion rate with an improvement of 3.82%. One of the causes of this improvement

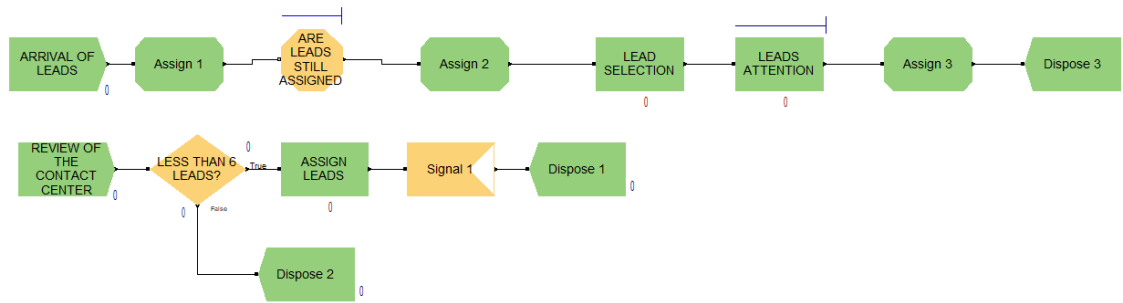


Figure 3: Simulation of the Improved Customer Service Process

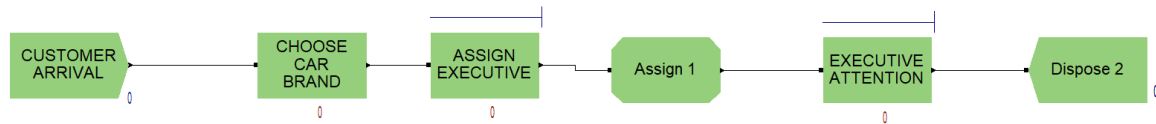


Figure 4: Simulation of the Improved Quotation Process

Measurement of the improvement problem								
Problem	Actual	Objective	Improved	Cause	Indicators	Actual	Objective	Improved
Low conversion rate of digital leads	0,92%	2%	3,82%	Excess customer volume	% Repeat customers	5%	1%	0,42%
				Repeat customers				
				Low skill level	% Digital leads	10%	50%	67,74%
				Extended process time	Total contribution time	37 min	20 min	16 min

Figure 5: Indicator Results

is the reduction of the percentage of repeat customers from 5% to 0.42%, achieving greater efficiency in the company's operations through Poka Yoke [10] [11]. The low skill level of salespeople was solved and measured with the percentage of digital leads that went from 10% to 67% thanks to the standardization of the profile, helping the competitiveness of the organization, having a social and economic impact [8]. Finally, the time in which a vehicle quotation is performed was reduced from 20 to 16 minutes with an improvement of 20% thanks to the application of BPM to detect, modify and eliminate processes that do not add value to the operation [13] [15].

6 CONCLUSION

The implementation of the proposed model in the automotive retail industry company, using the 4 tools, obtained results within the expected range. Regarding the proposed indicators, the 3.82% increase obtained in the conversion rate shows that the root causes of the problem were completely solved, so that the economic impact of \$393,916.87 analyzed in the objectives could be corrected,

obtaining a 15.73% improvement in the net profit obtained during the 9 months studied.

As for the other indicators, the 0.42% improvement in repeat customers demonstrates that the excess volume of customers, using Poka Yoke, substantially affects the time in which the EDVs attend to a quote, which obtained an improvement of 20%, and the skills of the salespeople, using profile standardization, which were measured by the percentage of digital leads they can attend to in a week, where an improvement of 67% was obtained, proving the effectiveness of the company's proposed commercial model and the potential for replication in other companies.

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